



Platypus milk could save lives

According to a research by Australian scientists, Platypus milk could help fight against antibiotic resistance. <http://digit.in/platypusmilk>



Nuking Asteroids

Russian scientists are working on a plan to nuke asteroids, by detonating miniature asteroids in labs. <http://digit.in/NukeAsteroids>

combine to create the complex network of ocean currents that is spread across the globe today.

WHY ARE THEY IMPORTANT

Back when traveling on the sea was not as advanced as today, and was the only option to travel across the world in (comparatively) shorter amounts of time, ocean currents played a huge role in speeding up travel around certain routes (and hindering on others). Circumnavigational expeditions harness ocean currents to this day to save time and fuel.

Most of the heat received by our planet is absorbed by the oceans. As a result, ocean currents, of days gone by and today, have persistently affected the climate of our planet. The absorbed heat is transported by currents across the globe, affecting climate everywhere. Take the Gulf Stream for example. It makes the climate of northwestern Europe much more temperate than any other place in the same latitude. That is because the Gulf Stream is a warm, swift Atlantic ocean current, and its northern offshoot, the North Atlantic Drift is what keeps Europe warmer than what it would be without it. The Gulf Stream is typically 100 kilometres (62 mi) wide and 800 metres (2,600 ft) to 1,200 metres (3,900 ft) deep, with

the maximum speed typically about 2.5 metres per second (5.6 mph). It has caused quite a few cyclones in the area.

These properties have a larger fallout. For instance, colder currents flowing in from the pole bring plankton rich waters with them, which are crucial for the survival of larger creatures in the sea.

TIP:

Ocean currents are measured in sverdrup (sv), where 1 sv is equivalent to a volume flow rate of 1,000,000 m³ (35,000,000 cu ft) per second.

Additionally, points where cold and warm currents meet are also rich in nutrients and hence usually filled with marine life. One of the most famous fishing centers in the world is Newfoundland in North

America, where the Labrador current and the Gulf Stream meet.

Another new area where ocean currents are showing a lot of promise is as an alternative source of energy. As opposed to wind and solar power, ocean currents are far more predictable and offer consistent, constant and clean energy. While the technology to harness this isn't as mature as other sources, it is still being explored around the world, with ideas such as submerged turbines.

WHERE THEY'RE HEADED

One would imagine that a phenomenon as disconnected from humans as earthly possible would be free from the clutches of climate change's vice. The rate at which sea ice and ice

around the poles is melting due to global warming also affects the formation of ocean currents. With a lower quantity of ice freezing, there's lower salt in the water at the poles. This leads to less water sinking to the depths, and less warm water rushing in to replace it - creating fewer, or less powerful ocean currents. Even in an indirect way, lesser ice leads to fewer winds, which is also a factor that adversely affects ocean currents.

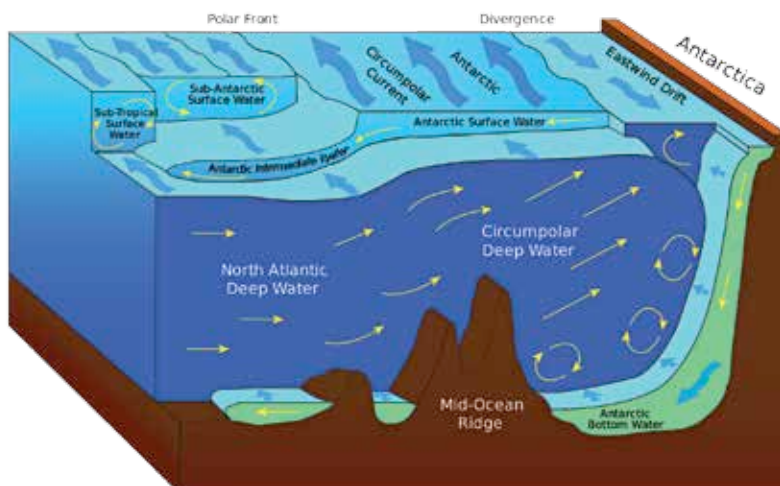
Due to the lack of circulation that ocean currents provide, there could be less oxygen in the water leading to a disruption of oceanic life cycles, and possibly the extinction of species that depend on oxygen in the water. The same would also lead to less CO₂ ab-



Deep Argo deployment

sorption by the ocean, leading to higher levels of CO₂ in the atmosphere. We all know what that can do.

Projects like Deep Argo are trying to further our understanding of deep ocean currents and how their gradual demise can be stopped. A submerged network of floats that look like a robotic, glass orb with an antenna to transmit signals to satellites and a two-meter long tail that detects when it has reached the seafloor is going down there to scan the unknown. Unlike previous Argo floats that worked on the surface, these can withstand the immense pressure of the deep for our bid to understand how the oceans have stored 90% of the heat received by Earth. This entire project will take some time to conclude. But, as with most issues that aid the cause of global warming on our planet, we need to act now before our oceans turn stagnant. 



A diagram with the various movements of water in the ocean described